

Orange(Yingzhi) Ao

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Education

Nanjing University	2021-2025
B.A., Financial Engineering	Ranking: 3/31 GPA: 92/100
Columbia University	2024
Exchange Student	GPA: 4.17/4.33

Relevant Courses

CS And Stats

**DS in Finance And Insurance [A+], **Predictive Modeling (in R), Statistical Machine Learning, Time Series, *Financial Engineering, *Artificial Intelligence, *Data Mining, *Options Investing Algorithms

Mathematics

**Stochastic Processes [A+], Numerical Analysis (in MATLAB), Discrete Mathematics [rank 1/23], ODE, *PDE, *Financial Mathematics

* is added before courses in progress; ** is added before grad-level courses.

Research Experiences

Multi-Period Compliance MFG-FBSDEs with DL	RA	2024.05 - Present
Columbia University Stats Department		

Under Prof. Steven Campbell’s supervision, I set up **Deep FBSDE Solvers** for the **multi-period compliance MFG** in REC markets. It was presented by him in the 2024 INFORMS Meeting. Next phase, more periods and/or inventory carryforward rules will be added to a more realistic model.

- **Inovation**. Extended the models to multi-period setting, renovated Deep FBSDE Solvers with smoothing and scaling layers and integrated numeric tricks (e.g. Sigmoid-Logit-Clamp) for greater stability.
- **Open-Source**. Developed user-friendly packages with detailed instructions, which facilitate experiments. All works have been documented in a report and made public in [my GitHub repository](#).

Insurance Pricing And Risk Modeling with ML	RA	2024.05 - 2024.08
Columbia University Actuarial Department		

Supervised by Carlos (Arocha Association), I novelly developed insurance pricing strategy with ML.

- **Innovatin**. Trained **K-Means** to identify risk clusters and customized **exposure rating** models for each cluster to price the insurance policies, which have not been implemented by the industry.
- **Professional Presenting**. Regularly delivered Statements of Work. **Communicated** the strategy with ML techniques in a professional manner accessible to both technical and non-technical clients.

Low-carbon Transition of China’s Power System	RA	2023.02 - 2023.08
Tsinghua-CTG Joint Center for Climate Governance and Low-carbon Transformation		

Supervised by the lead researcher Yang Qu, I assisted the team in composing the research report series "Low-Carbon Transition of China’s Power System". My key responsibilities and contributions are:

- **Co-Authorship**. Co-authored sections and assisted in drafting and reviewing to uphold report quality.
- **Data Support**. Gathered and visualized data throughout the series to enhance data-driven insights.

Project Experiences

Pair-Trading Optimization	Independent Researcher	2023.08 - 2023.09
Innovated a double-bound Bollinger strategy which significantly reduced fluctuations and turnover rate. Then I applied Genetic Algorithms (GA) to optimize pair weights with customized fitness functions.		
WorldQuant Brain	Consultant	2023.09 - 2023.12
Secured a gold medal (11th place out of 200+ participants) in the Brain x NJU Alphathon 2023. Also contracted as a consultant to provide alpha ideas and received allowances.		

Internship Experiences

Higgs Assets	High-Freq Quant Researcher	2024.08 - Present
Developed a peer-grouping system by predicting correlations, which will be applied to boost the performance of model using multi-head attention in the next phase.		
TrexQuant Investment	Global Alpha Researcher	2024.06 - 2024.08
Developed alphas based on studies in behavioral finance. Applied regression and clustering techniques.		
Chongwen Quantitative	Quant Strategy Researcher	2023.09 - 2024.02
Adapted and optimized the Whitney George Strategy to the CHN stock market. Built a trading simulation to generate instructions for traders. It's been actively used for trading since 2024 Q2.		

Skills

Programming
Deep Learning with PyTorch, markdown, L ^A T _E X Git commands. Basic R, C++, and MATLAB.
English Proficiency
111/120 in TOEFL and 333/340 in GRE, with overseas experience.
Writing and Typesetting
Academic and professional reports, statements of work, and presentations.
Collaboration And Communication
Skilled in providing constructive feedback and adaptable in a team through teamwork experience.